Section 5.2 and 5.3: Future and Present Value of an Annuity

Finite Mathematics MTH 132 Spring 2014

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TOTAL:

$$\sum_{n=0}^{5} 100(1.08)^{n-1} = 5866.60$$

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Future Value of an Annuity

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Sinking Fund Payment

$$R = \frac{S \cdot i}{(1+i)^n - 1}$$

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Example 5.2 and 5.3.3.

You need to save \$15,000 in 10 years. What size monthly payments should you make into an account which pays %8 compounded monthly? (payments at end of month)

5.3 Present Value of an Annuity; Amortization

Example 5.2 and 5.3.4.

Your college fund pays \$5,000 every every for 4 years. If the funds are in an account which accrues %5 compounded yearly, how much was originally in the account?

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yr 1: $17,729.75(1.05) - 5000 = 13616.23$

yr 2: $13616.23(1.05) - 5000 = 9297.05$

yr 3: $9297.05(1.05) - 5000 = 4761.90$

yr 4: $4761.90(1.05) - 5000 = 0$

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month 1: $219,780.9$

month 2: $219,560.7$

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month 12: $217,298.18$ (after \$15,828.12 in payments)

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- Amount owed:

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month 358: $2,618.37$

month 359: $1,312,45$

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Present Value of an Annuity

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Example 5.2 and 5.3.7.

Your college fund pays $5,000 every every 6 months for 4 years. If the funds are in an account which accrues %5 compounded semiannually, how much was originally in the account?